QP Code **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) MODEL QUESTION PAPER
SEMESTER V
DSC - MG5DSCZGY302, MG5DSCZIM301 - FUNDAMENTALS OF GENETICS
(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
MCQ
Answer all questions
Each question carries 1 mark

1. In Aa × Aa, what is the expected genotypic ratio? [A] / [CO1]
   (a) 1 : 2 : 1 (b) 3 : 1
   (c) 1 : 1 (d) 9 : 3 : 3 : 1

2. In codominance, a heterozygote shows: [U] / [CO1]
   (a) Both alleles expressed (b) Intermediate phenotype
   (c) Dominant allele only (d) Recessive allele only

3. The genotype of blood group AB is: [K] / [CO1]
   (a) I^AI^A (b) I^AI^B
   (c) I^Bi (d) ii

4. Complete linkage produces: [U] / [CO2]
   (a) Parental types only (b) Equal four classes
   (c) Recombinants only (d) More recombinants

5. In the ZW-ZZ system, the heterogametic sex is: [K] / [CO2]
   (a) Female, ZW (b) Male, ZZ
   (c) Female, ZZ (d) Male, ZW

6. A mutation arising in a germ cell is a: [K] / [CO3]
   (a) Germinal mutation (b) Somatic mutation
   (c) Structural mutation (d) Numerical mutation

7. The condensed, less active chromatin is: [K] / [CO4]
   (a) Heterochromatin (b) Euchromatin
   (c) Nucleosome (d) Chromatid

8. Which order shows increasing chromatin packing? [U] / [CO4]
   (a) DNA → nucleosome → solenoid (b) Solenoid → DNA → nucleosome
   (c) Nucleosome → DNA → solenoid (d) DNA → solenoid → nucleosome

9. Trisomy 21 causes: [K] / [CO5]
   (a) Down syndrome (b) Edward syndrome
   (c) Turner syndrome (d) Cri du chat syndrome

10. A karyotype 47,XXY indicates: [U] / [CO5]
    (a) Klinefelter syndrome (b) Turner syndrome
    (c) Down syndrome (d) Cri du chat syndrome

[1 x 10 = 10]

Part B
Short Answer Questions
Answer any 10 questions
Each question carries 3 marks

11. State Mendel's law of segregation and law of independent assortment. [K] / [CO1]

12.	Distinguish a test cross, a back cross and a reciprocal cross by stating the type of mating involved in each.	[U] / [CO1]
13.	An F2 ratio of 9:7 is obtained in a dihybrid cross. Identify the type of gene interaction and explain the basis of this ratio.	[A] / [CO1]
14.	Define sex-limited, sex-influenced and holandric inheritance.	[K] / [CO1]
15.	Name the type of sex determination in honey bees and state the ploidy of males and females.	[K] / [CO2]
16.	Explain the role of the SRY gene in human gonadal development.	[U] / [CO2]
17.	A somatic nucleus contains two X chromosomes. Predict the number of Barr bodies and justify your answer using the Lyon hypothesis.	[A] / [CO2]
18.	Name any two structural chromosomal aberrations and one numerical chromosomal aberration.	[K] / [CO3]
19.	Explain how a tautomeric shift can result in a base substitution mutation.	[U] / [CO3]
20.	Explain how addition or deletion of nucleotides can produce a frame-shift mutation.	[U] / [CO3]
21.	Differentiate polytene and lampbrush chromosomes with reference to their occurrence and one structural feature.	[U] / [CO4]
22.	Briefly explain the genetic basis of sickle cell anemia, cystic fibrosis and Tay-Sachs disease.	[U] / [CO5]

[3 x 10 = 30]

Part C

Short Essay Questions
Answer any 6 questions
Each question carries 5 marks

23.	A dihybrid F1 plant with genotype RrYy is test-crossed with rryy. Assuming independent assortment, predict the phenotypic classes and their expected ratio. State the Mendelian principle illustrated.	[A] / [CO1]
24.	Explain polygenic inheritance and pleiotropism with one suitable example each. Briefly state the role of modifying genes.	[U] / [CO1]
25.	In a two-point test cross, the progeny are AB = 410, ab = 390, Ab = 100 and aB = 100. Identify the parental and recombinant classes, calculate the recombination frequency and determine the map distance between genes A and B.	[A] / [CO2]
26.	Explain the genic balance theory of sex determination in Drosophila. How does the X:A ratio determine male, female and intersex conditions?	[U] / [CO2]
27.	Describe the CIB method used to detect X-linked recessive lethal mutations in Drosophila.	[U] / [CO3]
28.	List the standard pedigree symbols and conventions for male, female, affected individual, carrier, mating and line of descent.	[K] / [CO4]
29.	Explain the genetic basis of phenylketonuria, alkaptonuria and albinism.	[U] / [CO5]
30.	List the major objectives of the Human Genome Project. Define genetic counselling, eugenics and euthenics.	[K] / [CO5]

[5 x 6 = 30]